



CONTROL SYSTEMS

Faculty team

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CONTROL SYSTEMS

Introduction



Contents



Unit 1: Mathematical Models of Systems	Transfer Function Models: Differential Equations of Physical Systems. Block Diagram Models. Signal Flow Graph Models. State Variable Models: The State Variables of a Dynamic System The State Differential Equation; Signal Flow Graph State Models; Transfer function from the state equation; The Time Response and the State Transition Matrix.
Unit 2: The Performance of Feedback Control Systems of a system	Feedback Control System Characteristics: Open and Closed Loop Control Systems; Sensitivity of Control Systems to Parameter Variations; Disturbance Signals in a Feedback Control System; Cost of feedback. Performance of a second-order system: Test Input Signals, The s-plane root location and the Transient Response, steady-state errors of unity and non-unity feedback systems. The Stability of Linear Feedback Systems: Concept of stability; Routh–Hurwitz stability criterion; relative stability of feedback control systems; stability of state variable systems

Contents

Unit3: Stability analysis of a system & Frequency Domain Analysis	The Root Locus Method: Concept, Procedure, Design utilizing the Root Locus method. Frequency Response Measurements: Performance specification in the frequency domain. The Bode Diagram, Time-Domain Performance Criteria Specified in the frequency Domain, System Bandwidth, Relative Stability
Unit 4: Design of Feedback Control Systems	Stability in the Frequency Domain: Mapping Contours in the s-plane; The Nyquist Criterion; and the Nyquist Criterion. System Design: Cascade Compensation Networks, Phase-lead and phase-lag design using Bode diagram, P, PI, PID controllers, PID tuning. Design of State Variable Feedback Systems: Controllability; Observability

Study Material

Book Type	Author & Title	Edition	Publisher	Year
Text 1	<i>“Modern Control Systems”</i>, Richard C. Dorf, Robert H. Bishop	14th	Pearson	2021 Impression
Ref 1	<i>“Control Systems Engineering”</i> , I. J. Nagrath, M. Gopal	7 th	New Age	2021
Ref 2	<i>“Modern Control Engineering”</i> , K. Ogata	5 th	Pearson	2019 Impression
Ref 3	<i>“Automatic Control Systems”</i> , B C Kuo	9 th	Prentice Hall	2014

Draft Assessment Plan



Event		Marks	Mode
ISA	ISA 1 (Unit 1 &2)	40 reduced to 20	HYBRID
	ISA 2 (Unit 3 &4)	40 reduced to 20	
Projects/ Assignment	A/H/P-1	4	Pen and Paper/ Simulation/ Presentations
	A/H/P-2		
	A/H/P-3 A/H/P-4	6	
	Total	10	
Total ISA		50	
ESA	Unit 1 till Unit 4	100 Reduced to 50	HYBRID
Total ISA + ESA		100	

- **A system is..**
 - A set of things working together as parts of a mechanism or an interconnecting network to perform a function.
 - An interconnection of elements and devices for a purpose.
 - **Ex, Bus, two wheeler, fan etc.,**

CONTROL SYSTEMS

What is a control system?

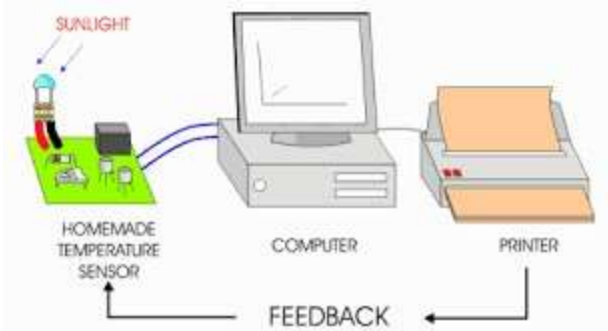
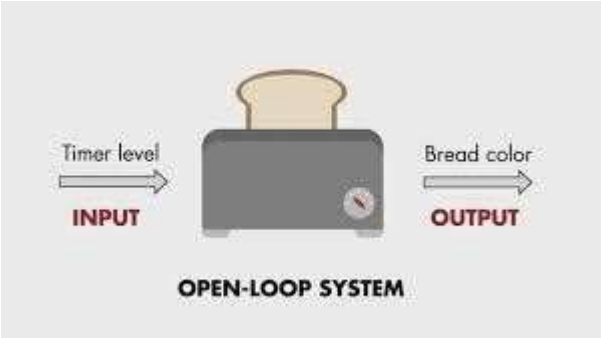


- A combination of components that act together to provide a desired system response.
- A **control system** manages, commands, directs, or regulates the behavior of other devices or **systems** using **control** loops. It provides the desired response by controlling the output.
- The control system maintain the actual system performance close to a desired set of performance specifications.



CONTROL SYSTEMS

Examples of Systems



A SIMPLE EXAMPLE OF CONTROL TECHNOLOGY:

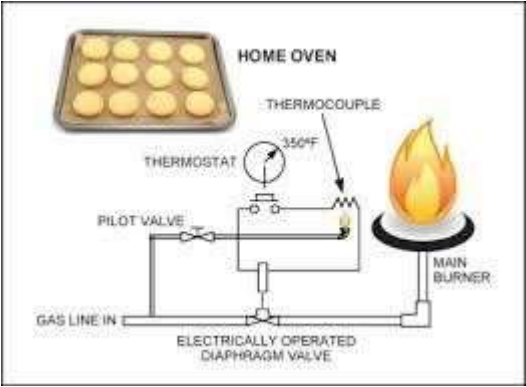
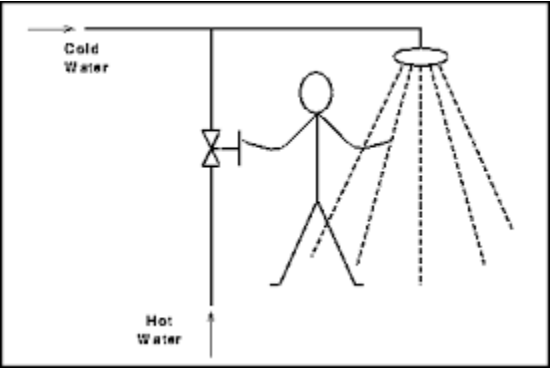
Traffic Lights

- **GOAL:** To be able to define what control technology is.
- **GOAL:** To be able to identify everyday applications and give real-life examples that involve control technology.
- **GOAL:** Understand the need for instructions to be in order.

Example of Control Systems



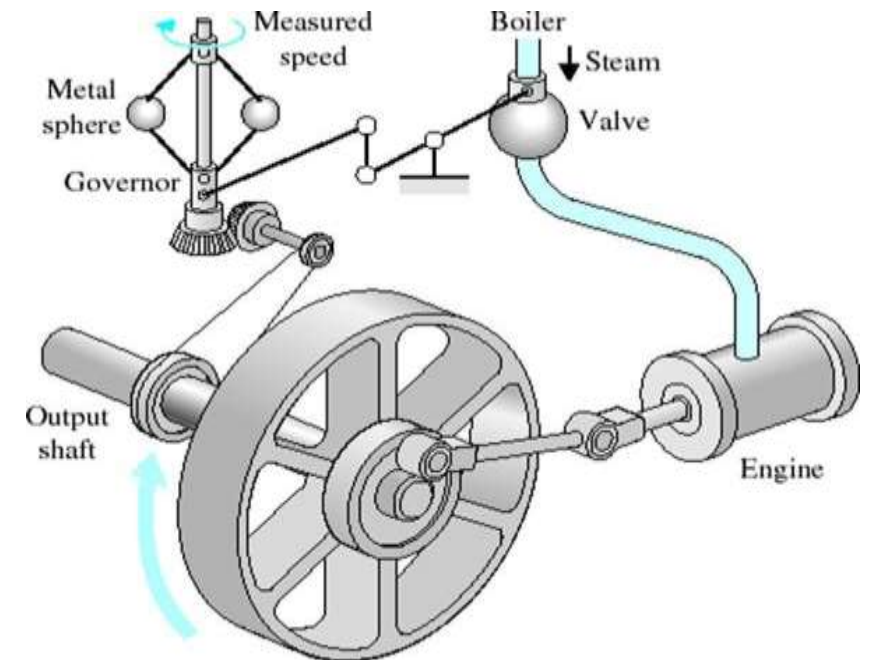
A manual control system for regulating the level of fluid in a tank by adjusting the output valve. The operator views the level of fluid through a port in the side of the tank.



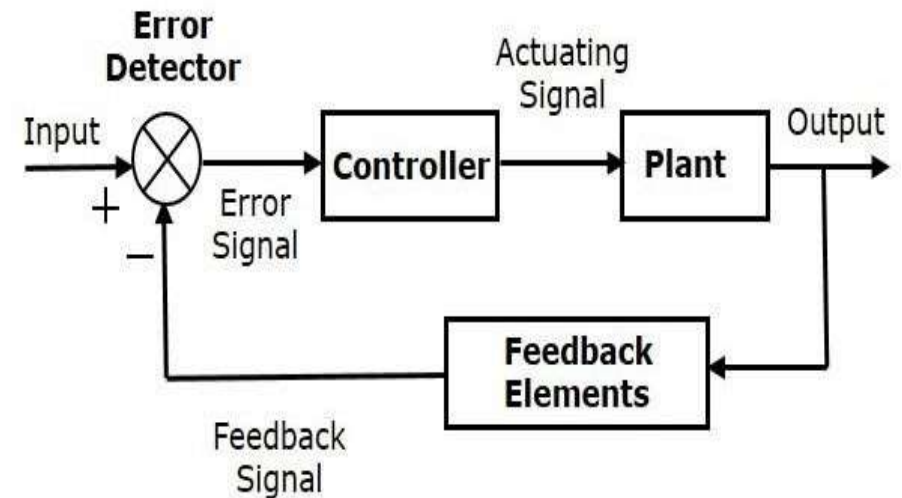
CONTROL SYSTEMS

Why do we need control?

- Speed regulation(ex, Watt's Flyball governor, 1788) and maintain system response time
- Reduces the effect of variations in load(ex, Power systems)
- Maintain precision and robust(ex, Missile, Robotic arm)
- To handle huge and complex operations(ex, Airbus-A380)
- To ensure maneuverability and agility(ex, Fighter Aircraft)



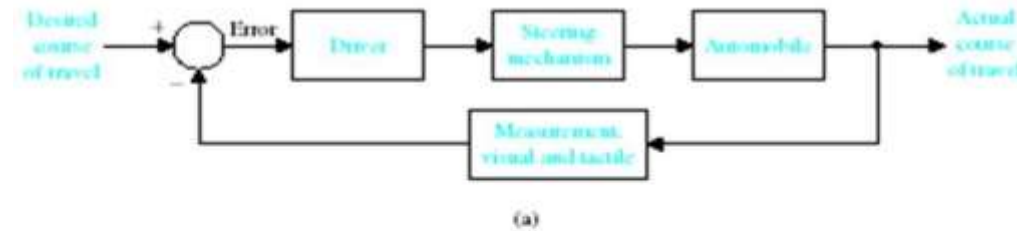
- The Plant or process or system to be controlled
- The *sensor* is a measurement device for measuring the output of the system.
- The comparator and the controller determines the corrective action by comparing the actual output of the plant and the expected output/ reference.
- The actuator takes the corrective action to return the system to its expected output.



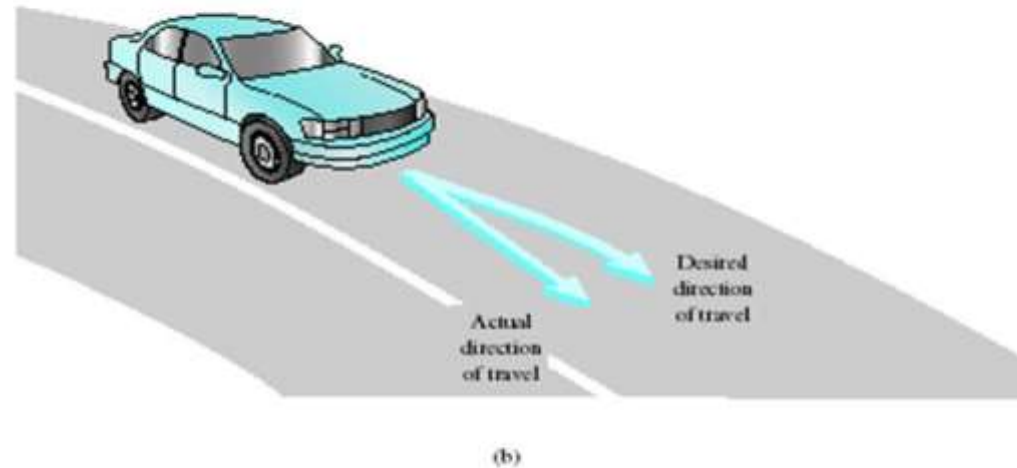
CONTROL SYSTEMS

Examples of Control Systems

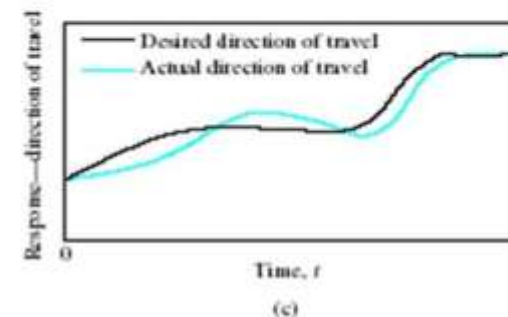
(a) Automobile steering control system.



(b) The driver uses the difference between the actual and the desired direction of travel to generate a controlled adjustment of the steering wheel.



(c) Typical direction- of-travel response.



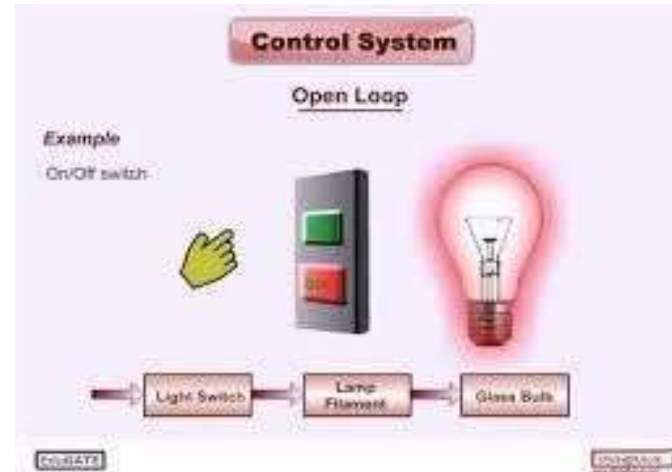
CONTROL SYSTEMS

Classification of control systems

- Open Loop



Open Loop Control System



- Systems in which the output has no influence or effect on the **control** action of the input signal , i.e output signal or condition is neither measured nor “fed back” for comparison with the input signal

CONTROL SYSTEMS

Classification of control systems

- **Why Systems fail?**



How do you avoid Bread Blackening?

Causes: Timer failure (Hard Setting of Controller),
Excess timer setting [Open Loop Control System]

Solution: Bread Color Sensing. Use this measure
to exercise restraint on turning on/off the heater.
[Closed Loop or Feedback System]



What failed here?

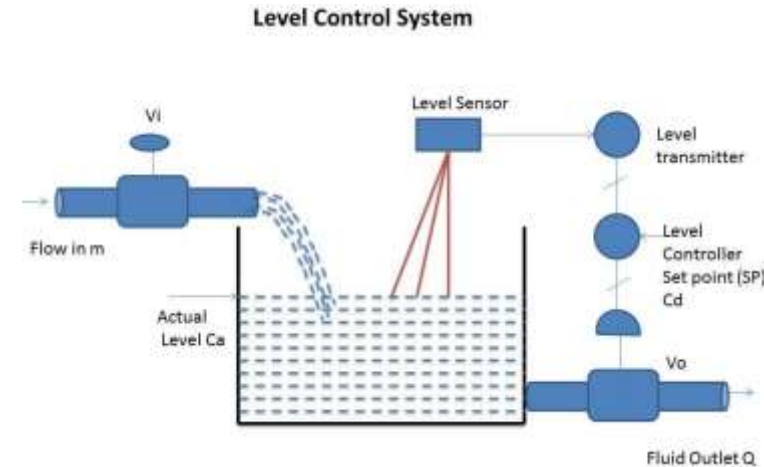
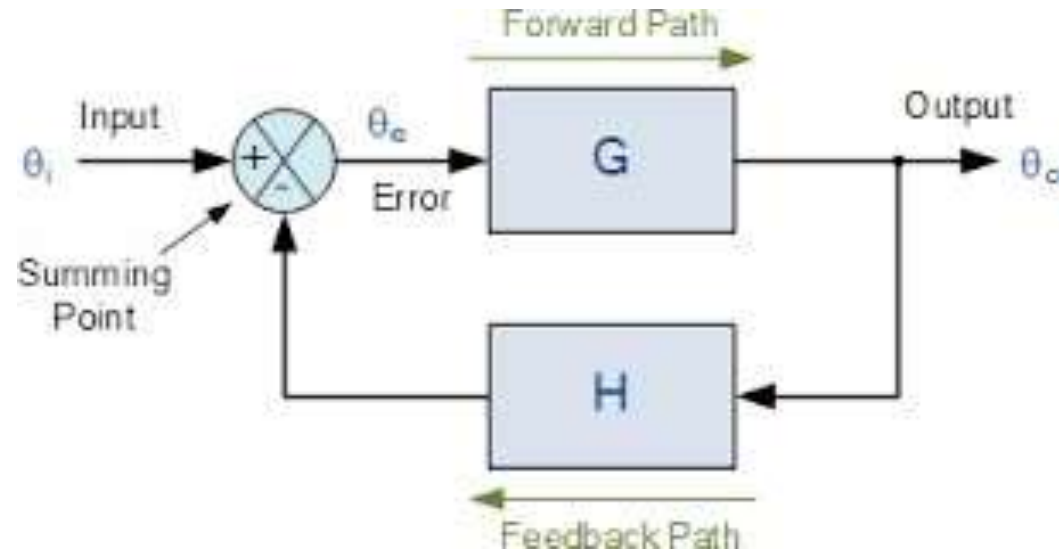
Cause: Depth not visible

Solution: Depth Sensing

CONTROL SYSTEM

Classification of control systems

- Closed Loop



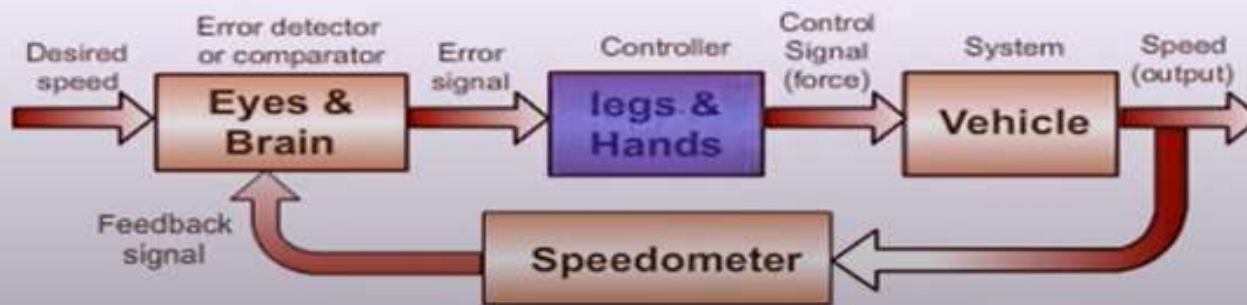
A **closed loop control** system considers the current output and alters it to the desired condition. The **control** action in these systems is based on the output.

CONTROL SYSTEMS

Classification of control systems

Close Loop System

Example



Find the error in the block diagram

[Open Loop and Closed Loop Control Systems - Differences, Examples - YouTube](#)



THANK YOU

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